

TYNEMOUTH, United Kingdom

WMO: 032620

Lat: 55.017N

Lon: 1.417W

Elev: 98

StdP: 14.64

Time Zone: 0.00 (EUW)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.3	31.6	23.9	18.2	32.8	25.9	20.0	34.6	51.6	42.1	47.9	40.2	14.9	280	0.805

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	69.8	61.4	67.4	60.5	65.4	59.2	63.7	67.5	62.0	65.6	60.6	64.1	14.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	83.5	65.2	60.4	78.8	63.9	58.8	74.3	62.7	29.0	67.7	27.8	66.0	26.8	64.2	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 46.3	39.7	33.4	DB	24.4	77.0	3.7	5.2	21.7	80.7	19.5	83.8	17.4	86.7	14.7	90.5	(4)
(5)			WB	23.8	66.8	3.7	2.7	21.1	68.7	18.9	70.3	16.8	71.8	14.0	73.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	40.9	41.0	43.5	44.8	50.1	54.1	58.9	58.9	55.0	50.5	45.1	42.5
	DBStd	7.89	4.88	5.46	4.89	4.75	4.36	4.17	3.46	4.12	4.01	4.74	4.65	5.06
	HDD50	1431	283	255	210	171	53	9	0	0	6	52	155	237
	HDD65	5909	746	671	668	606	461	327	191	194	302	450	596	697
	CDD50	1006	2	4	8	15	57	132	277	276	155	66	9	5
	CDD65	9	0	0	0	0	0	0	3	5	1	0	0	0
	CDH74	15	0	0	0	0	0	0	4	9	2	0	0	0
	CDH80	2	0	0	0	0	0	0	0	2	0	0	0	0
Wind	WSAvg	15.8	19.1	18.8	17.6	16.5	14.0	13.4	12.6	13.2	14.6	15.9	16.6	17.3
Precipitation	PrecAvg	28.8	2.4	2.0	1.7	2.1	2.0	2.4	2.5	2.7	2.3	2.8	3.2	2.7
	PrecMax	40.5	6.5	4.5	4.0	5.8	3.9	6.5	6.1	6.2	5.8	6.3	6.9	5.9
	PrecMin	21.3	0.7	0.3	0.4	0.3	0.6	0.6	0.4	0.5	0.7	0.7	1.1	1.2
	PrecStd	4.6	1.3	1.1	0.9	1.6	0.9	1.5	1.3	1.2	1.3	1.1	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	56.5	58.5	61.3	65.0	68.4	72.9	73.6	71.1	64.3	57.8	57.9
		MCWB	50.8	52.2	52.0	54.7	56.2	60.0	63.8	63.3	63.6	58.7	53.6	54.4
	2%	DB	52.8	54.1	55.2	57.7	61.6	65.1	69.7	69.6	65.7	61.9	55.8	54.3
		MCWB	49.1	50.0	49.6	52.1	54.4	58.1	61.4	61.7	59.4	57.3	52.2	51.3
	5%	DB	50.5	52.1	53.1	55.1	58.8	62.7	67.1	67.4	63.1	59.4	53.8	52.1
		MCWB	47.3	48.4	48.7	50.5	53.0	57.0	60.2	60.7	57.2	55.4	50.8	49.1
	10%	DB	48.1	49.6	51.0	52.3	56.6	60.6	64.9	65.5	61.1	57.2	51.8	49.9
		MCWB	45.3	46.4	47.0	48.4	52.2	55.7	58.7	59.9	56.1	53.4	49.3	47.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	52.9	52.7	56.3	58.1	62.1	65.4	66.0	64.8	59.8	54.4	54.9
		MCDB	54.6	56.0	57.2	60.4	62.8	67.0	71.5	70.8	69.9	63.8	57.0	58.0
	2%	WB	49.5	50.4	50.4	52.9	55.6	59.6	62.8	64.1	60.9	57.9	52.8	51.5
		MCDB	52.6	53.5	54.2	56.5	59.8	62.8	67.8	67.5	64.4	61.2	55.1	53.9
	5%	WB	47.5	48.7	49.1	50.9	53.9	58.1	61.3	62.6	58.7	55.9	51.3	49.3
		MCDB	50.1	51.7	52.6	54.5	57.6	61.6	65.3	65.4	61.6	58.7	53.3	51.7
	10%	WB	45.7	46.6	47.5	48.9	52.6	56.5	60.1	61.1	57.2	54.0	49.7	47.3
		MCDB	48.0	49.4	50.6	52.0	56.1	60.1	63.6	64.4	59.8	56.0	51.5	49.6
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.9	8.1	7.4	7.3	7.0	8.1	8.5	8.4	7.6	6.6	6.5
		MCDBR	9.5	8.7	10.8	12.0	11.9	11.9	12.4	12.0	12.0	10.4	8.6	8.7
		MCWBR	9.1	7.3	8.2	8.6	7.2	6.5	7.3	6.7	8.0	8.0	8.0	8.1
	5% WB	MCDBR	9.5	8.7	10.5	11.7	10.8	10.0	10.7	9.6	11.0	9.5	7.9	8.7
		MCWBR	9.1	7.7	8.5	8.9	7.2	6.8	7.2	6.5	8.7	8.0	7.5	8.3
Clear-Sky Solar Irradiance	taub	0.318	0.335	0.366	0.390	0.387	0.390	0.391	0.382	0.362	0.345	0.319	0.302	
	taud	2.321	2.388	2.319	2.277	2.340	2.359	2.362	2.401	2.458	2.458	2.377	2.287	
	Ebn at Noon	193	236	254	264	272	272	269	264	255	230	193	170	
	Edh at Noon	17	23	31	37	36	36	36	32	27	22	16	15	
All-Sky Solar Radiation	RadAvg	180	381	744	1211	1569	1573	1505	1253	891	484	227	133	
	RadStd	23	42	93	116	82	125	117	82	75	39	20	14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page